

AMENDMENT NO. 4 MARCH 1982

TO

IS : 2825-1969 CODE FOR UNFIRED PRESSURE VESSELS

Alterations

- | <i>Sl No.</i> | <i>Sl No.</i> |
|---|--|
| 1. [<i>Sl No. 26 of Amendment No. 2 to IS : 2825-1969 dated October 1978, new clause 6.3.2(a), 4th line</i>] — Substitute 'hole' for 'whole'. | 4. (<i>Page 5, clause 1.1.1</i>) — Substitute the following for the existing clause:

'1.1.1 This code covers minimum construction requirements for the design, fabrication, inspection, testing and certification of fusion-welded unfired pressure vessels including those intended for transportation in ferrous as well as in non-ferrous metals.' |
| 2. (<i>Sl No. 24 of Amendment No. 3 to IS : 2825-1969 dated July 1979, new Fig. G.72</i>) — Substitute ' t_2 ' for ' t_3 ' on the right hand side of the Fig. Also under 5th line of Note (e) substitute '2 mm' for '2.9 mm'. | 5. [<i>Page 39, clause 3.12.1(c)</i>] — Add the following new clause after this clause:
d) A typical design of jacketed portion is covered under Appendix P. |
| 3. [<i>Page 6, clause 1.2.5.2(f)</i>] — Add the following new clause after this clause:

'g) Causes such as acceleration, deceleration, etc, in the case of vessels intended for transportation.' | 6. (<i>New Appendix P</i>) — Add the following new Appendix P at the end of the code: |

' APPENDIX P

[Clause 3.12.1(d)]

DESIGN OF JACKETED PORTION OF VESSEL

P-1. SCOPE

P-1.1 The rules in this appendix cover minimum requirements for the design and construction of the jacketed portion of the vessel. The jacketed portion of the vessel under analysis is defined as the inner and outer walls, the closure devices and other parts within the jacket which are subject to pressure stresses.

P-1.2 For the purpose of this part jackets are assumed to be integral pressure chambers, attached to vessels for one or more purpose such as:

- a) to heat the vessel and its contents,
- b) to cool the vessel and its contents, and
- c) to provide a sealed insulation chamber for the vessel.

P-1.3 This appendix does not cover rules to cover all details of design and construction. These rules are, therefore, established to cover most common type of jackets but are not intended to limit configurations to those illustrated or otherwise described therein.

P-1.4 All other parts of this code shall apply unless otherwise stated in this appendix.

P-1.5 Typical jacketed connections are given in Fig. G.45 to G.66 of Appendix G.

P-2. TYPE OF JACKETS AND NOTATION

P-2.0 Jacketing of process vessels is usually accomplished by use of one of the following main available types:

- a) Conventional jackets (*see* Fig. P.1 and P.2);
- b) Half pipe coil jackets or limpet coil jackets (*see* Fig. P.3, P.4 and P.7);
- c) Dimple jackets (*see* Fig. P.6); and
- d) Heater channels type jackets (*see* Fig. P.5).

P-2.1 Conventional Jacket (*see* Fig. P.1 and Fig. P.2)

- a) The normal configuration of conventional jacket is as shown in Fig. P.1, Type A. This assures the most efficient heat transfer to the maximum surface area of the vessel.
- b) An often used variation of this configuration is made (*see* Fig. P.1, Type B) by dividing the straight side into two or more separate jackets.

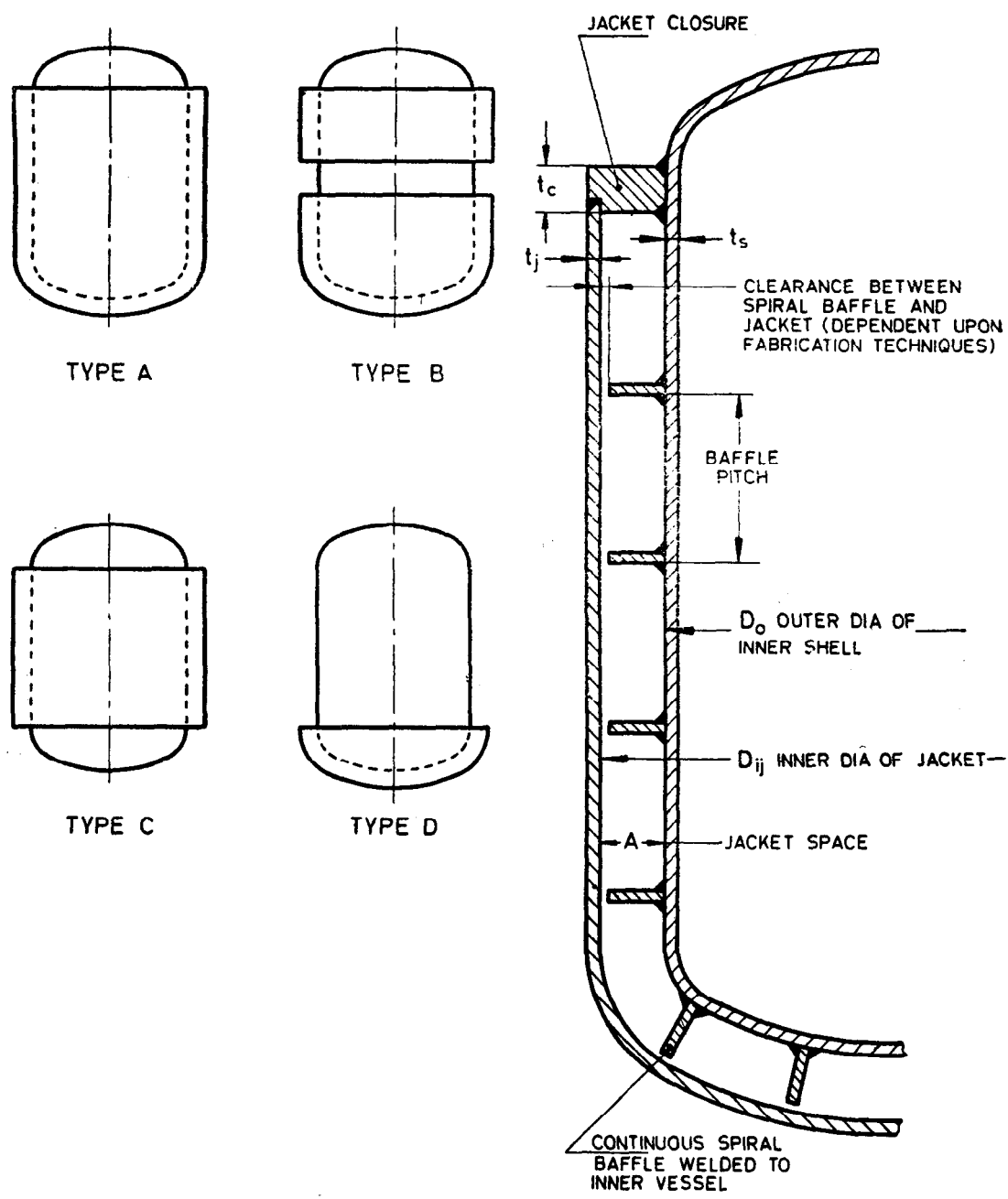


FIG. P.1 CONVENTIONAL JACKET IN CROSS SECTION

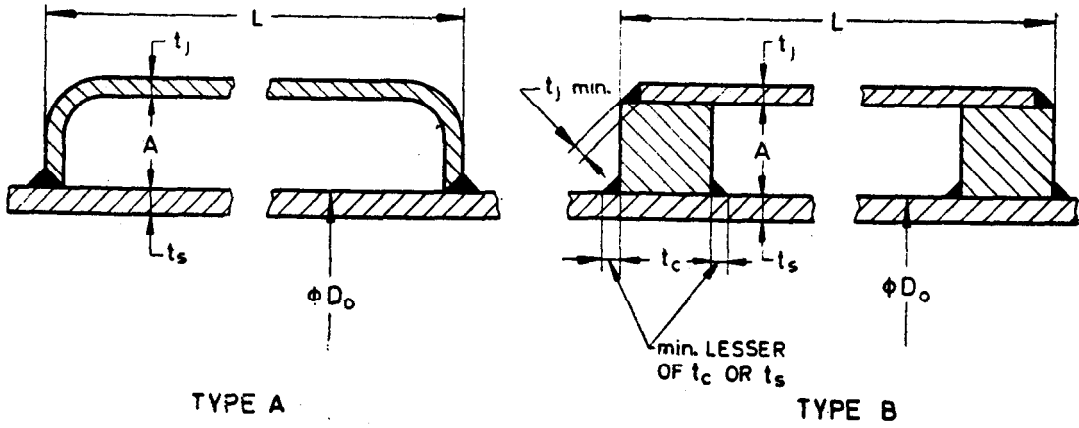


FIG. P.2 CONVENTIONAL JACKET

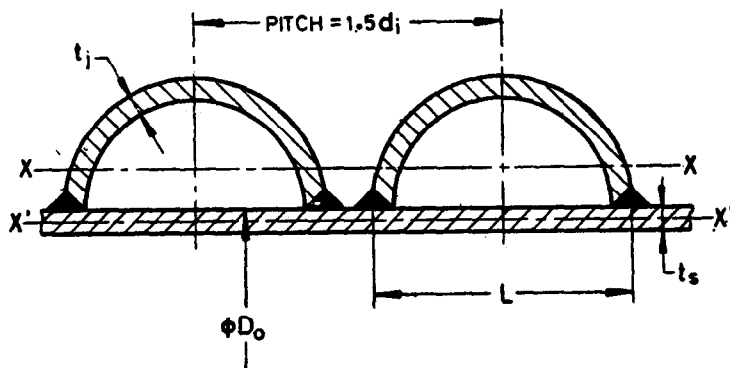


FIG. P.3 HALF PIPE COIL JACKET

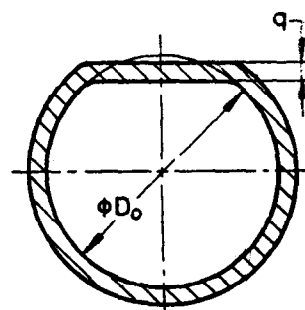


FIG. P.4 FLATNESS SECTION

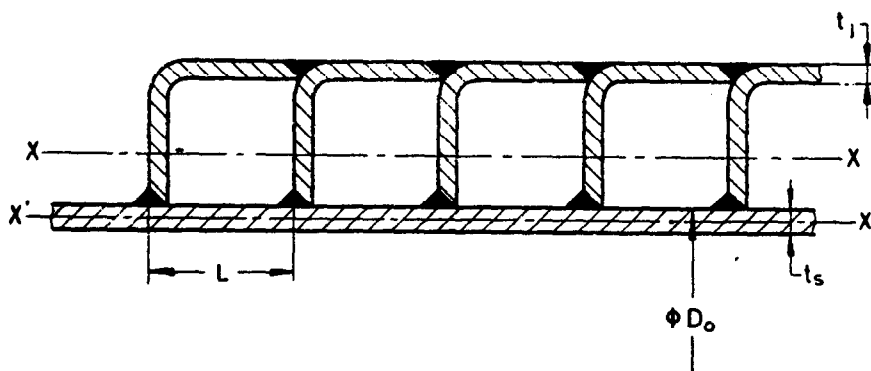


FIG. P.5 HEATER CHANNEL TYPE JACKET

c) If desired the vessel can be jacketed on the straight side on varying from complete to partial vertical coverage (see Fig. P.1, Type C); and

d) A jacket can also be fabricated to cover the bottom head only (see Fig. P.1, Type D).

P-2.2 Half Pipe Coil Jackets or Pimpet Coil Jackets (see Fig. P.3 and Fig. P.7) — The half pipe coil jacket is especially recommended for high temperature services. Because there are no limitations to the number and location of inlet and outlet connections, the half pipe coil jacket can be divided into multiple zones (see Fig. P.7) for maximum flexibility and efficiency. The half pipe coil design usually allows reduction in thickness of inner wall of the vessel.

P-2.3 Dimple Jacket (see Fig. P.6) — The design of dimple jacket permits construction from light gauge metals without sacrificing the strength required to withstand specified pressures. Manifold should be designed to avoid stress concentration due to discontinuity stresses and flexible hoses should be used to eliminate all external forces on the jacket connections and their manifolds.

P-2.4 Notation

Symbol	Unit	Description
A	mm	Jacket space.
D_1	mm	Inner diameter of shell.
D_o	mm	Outer diameter of shell.
D_{1j}	mm	Inner diameter of the jacket.
d_1	mm	Inner diameter of half pipe coil.
E	kgf/mm ²	Modulus of elasticity of material at operating temperature.
f	kgf/mm ²	Maximum allowable stress value.
$\bar{J}$	—	Weld joint fact
L	mm	Design length jacket section.
l/m	—	Poisson's ratio.
p	kgf/cm ²	Maximum operating pressure in jacket.
P	kgf/cm ²	Maximum operating pressure in shell.
t_o	mm	Thickness of closure member.

Symbol	Unit	Description	Symbol	Unit	Description
t_j	mm	Thickness of outer jacket wall.	t_{rj}	mm	Required minimum thickness of outer jacket wall exclusive of corrosion allowance and manufacturing tolerances.
t_s	mm	Thickness of inner vessel wall.	t_{rs}	mm	Required minimum thickness of inner shell wall exclusive of corrosion allowance and manufacturing tolerances.
t_{rc}	mm	Required minimum thickness excluding corrosion allowance of the closure member.			

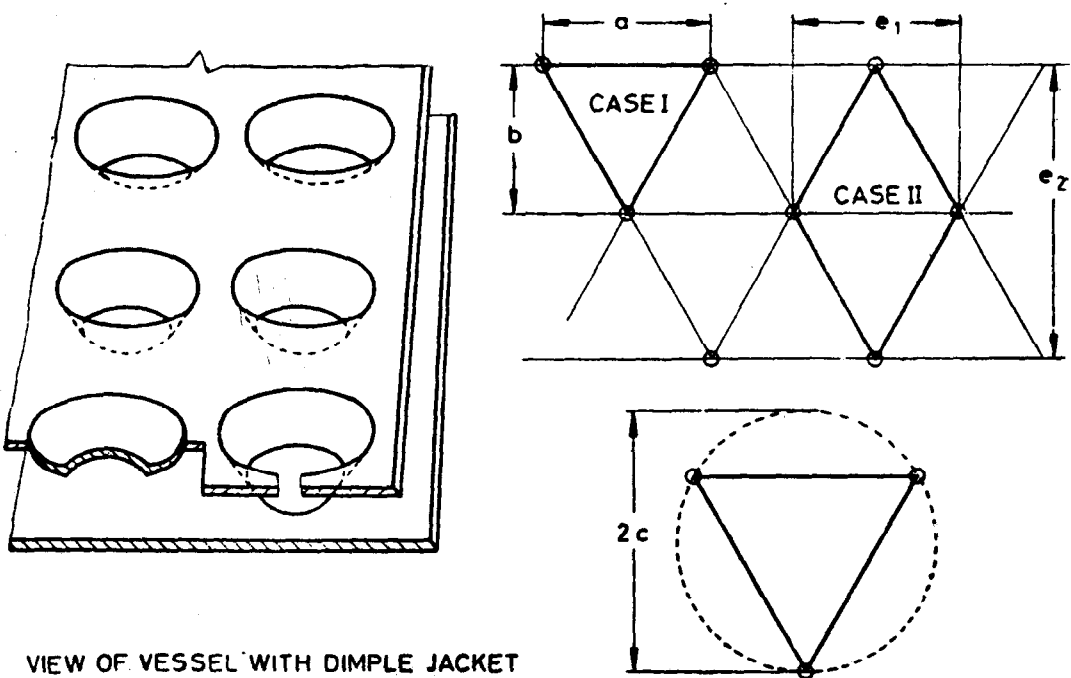
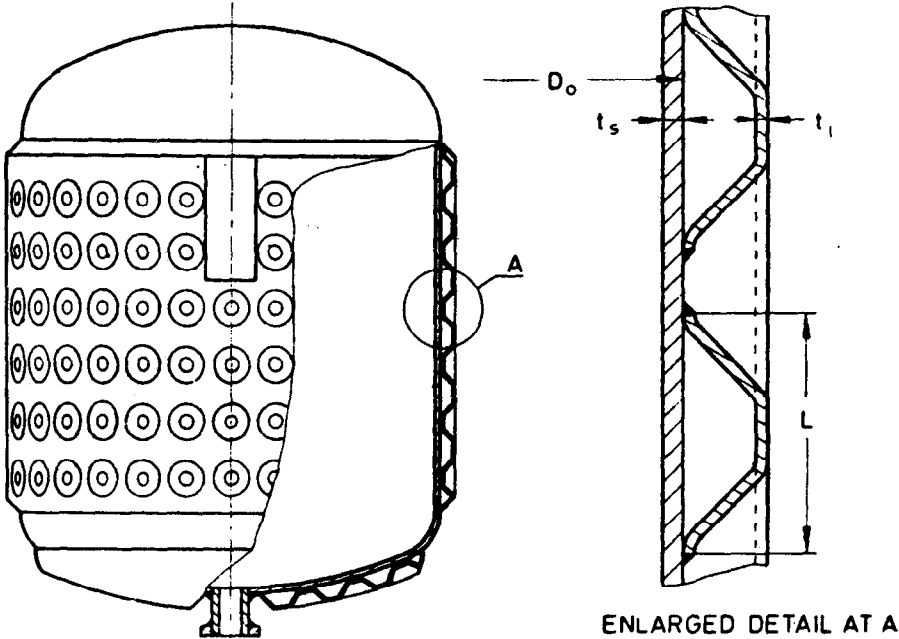


FIG. P.6 DIMPLE JACKET

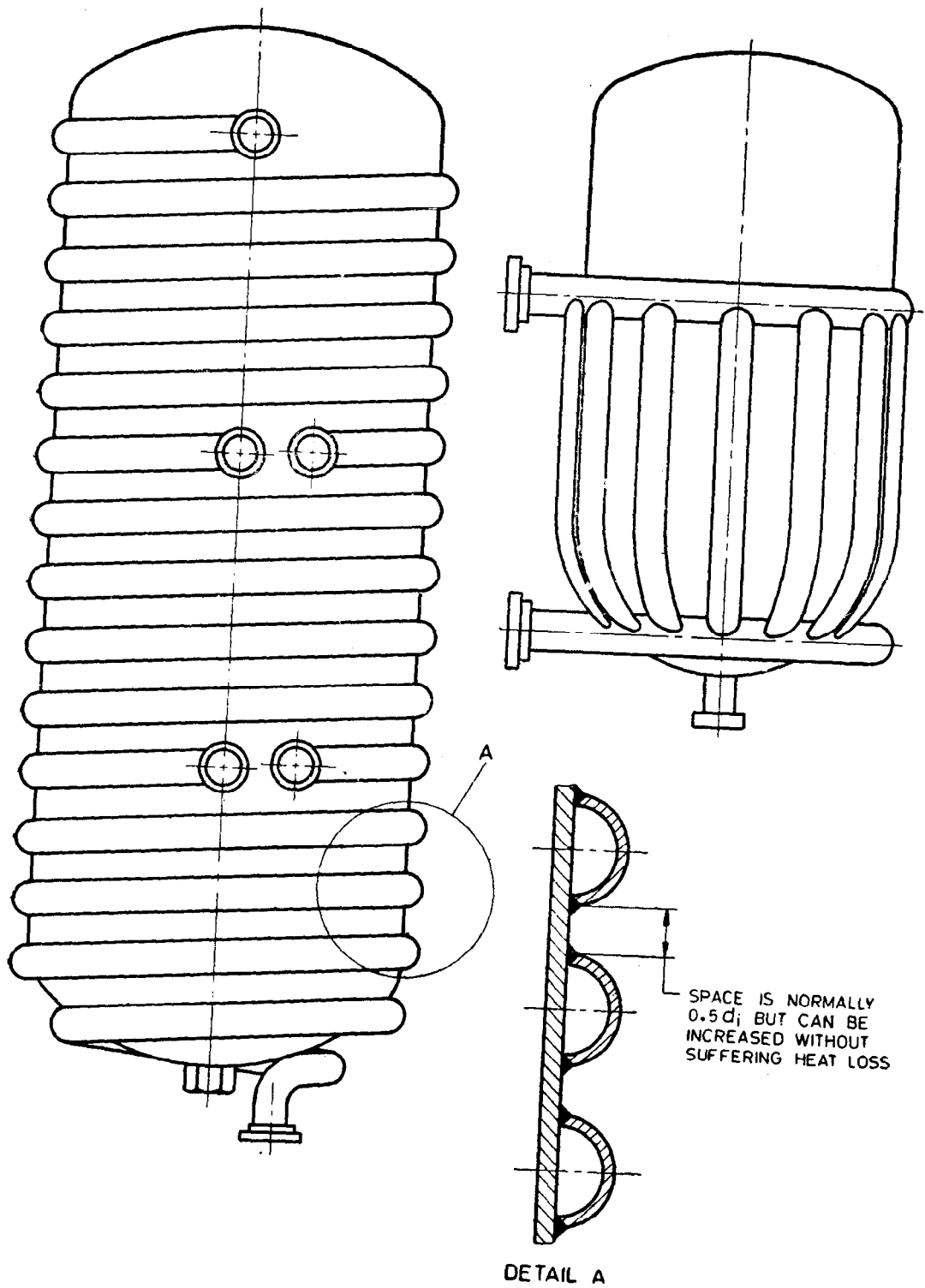


FIG. P.7 HALF PIPE COIL JACKET

P-3. DESIGN CRITERIA

P-3.1 Conventional Jacket

P-3.1.0 Design shall comply with the applicable requirements of the code except where otherwise provided for in the appendix.

P-3.1.1 Shell and head thickness shall be determined by the appropriate formulae given in Section 3 of this Code. In consideration of loadings particular attention to the effects of local internal and external loads and expansion differentials at the design temperature shall be given. Where vessel supports are attached to the jacket, consideration shall be given to the transfer of supported load of the inner vessel and contents.

P-3.1.2 The use of impingement plates or baffles at the jacket inlet connection to reduce erosion of inner wall shall be considered for media where vapours are condensed, that is, steam.

P-3.1.3 The width of jacket space shall not exceed the value given below:

$$A = \frac{400 \times f \times \bar{J} \times t_s^2}{p \times D_{ij}} - 0.5 (t_s + t_j) \quad \dots\dots (P.1)$$

P-3.1.4 Design for Jacket Closures

P-3.1.4.1 Closures of Type B of Fig. P.2 (Fig. G-55 modified) shall be used only for Type C of Fig. P.1 jacketed vessels and with further limitation that t_{rj} does not exceed 15 mm. The required minimum thickness for the closure bar shall be greater of the following:

$$t_{rc} = 2 (t_{rj}) \quad \dots\dots (P.2)$$

$$t_{rc} = 0.0866 \times A \times \sqrt{\frac{p}{f}} \quad \dots\dots (P.3)$$

P-3.1.4.2 Closure bar and closure bar to inner vessel welds of the type shown in Fig. G.49, G.50, G.51 may be used in any of the types of the jacketed vessels shown in Fig. P.1. The required minimum closure bar thickness shall be determined as below:

$$t_{rc} = 0.12247 \sqrt{\frac{p D_o A}{f}} \quad \dots\dots (P.4)$$

P-3.1.5 Minimum weld size requirements are as detailed in Fig. G.45 to G.59.

P-3.2 Half Pipe Coil Jackets

P-3.2.1 The foregoing analysis is based on the following assumptions which may be suitably modified in case of any variations.

- shell and half coil are made of the same metal,
- the half coil is semicircular,
- both shell and coil cylinders are considered as thin in relation to their diameters,

- the coils shall be pitched not less than $1\frac{1}{2}$ times d_1 , centres apart, and

NOTE — Very close spacing of coils is not recommended as it leads to bad weld joint, yields very little heat transfer benefit as the space in between the coils has an effect like fin on a finned tube.

- half coil takes no load in comparison with the shell.

P-3.2.2 Design of Half Coil

P-3.2.2.1 A half coil is a part of torus for which stresses have been analyzed by 'Timoshenko'. It can be shown that maximum hoop stress in the coil will occur at the junction with the shell and is

$$f_{c1} = \frac{p d_1}{200 \times t_{rj} \times \bar{J}} \quad \dots\dots (P.5)$$

and longitudinal stress is

$$f_{c2} = \frac{p d_1}{400 t_{rj} \times \bar{J} + 250 t_{rs} \times \bar{J}} \quad \dots\dots (P.6)$$

The half coil is, therefore, usually designed on a simple hoop stress basis.

P-3.2.3 Design of Shell

P-3.2.3.1 Total hoop stress — The total hoop stress in the shell, f_{st} is, sum of the hoop stress f_{s1} due to vessel pressure, and longitudinal stress f_{c2} in the coil caused by coil pressure.

$$f_{st} = \frac{P D_1}{200 t_{rs} \times \bar{J}} + \frac{p d_1}{400 t_{rj} \times \bar{J} + 250 t_{rs} \times \bar{J}} \quad \dots\dots (P.7)$$

P-3.2.3.2 Total longitudinal stress

- The total longitudinal stress in the wall is made of three factors:

- the longitudinal stress f_{s2} due to vessel pressure,
- the longitudinal stress f_{s3} due to coil pressure,
- the bending stress f_{s4} caused by distortion of the shell at the junction with the coil.

$$f_{s2} = \frac{P D_1}{400 \times t_{rs} \times \bar{J}} \quad \dots\dots (P.8)$$

$$f_{s3} = \frac{p d_1}{200 \times t_{rs} \times \bar{J}} \quad \dots\dots (P.9)$$

- The local bending stress in the shell due to the pressure in the coil should be assessed for the case when the internal pressure for the vessel is at its minimum, that is, atmospheric or vacuum. A simplified approximation based on 'Continuous Beam Theory' predicts a maximum bending moment, M_{max} in shell of:

$$M_{max} = \Delta P d_o^2 / 900 \quad \dots\dots (P.10)$$

Also

$$f_{s4} = \frac{2 \Delta P d_o^2}{300 t_{rs}^2} \quad \dots\dots (P.11)$$

(Negative value of $\Delta P = (p - P)$ need not be considered as this refers to high internal pressure. Bending moment due to internal pressure is insignificant compared with direct membrane forces which must have already been allowed for the hoop stress calculations.

$$f_{sl} = f_{s2} + f_{s3} + f_{s4} \quad \dots\dots (P.12)$$

- c) For critical analysis 'Cylindrical Shell Theory' concept may be followed.

P-3.2.4 Maximum Equivalent Stress at Coil to Shell Junction

The metal at the junction of coil to shell is subject to all individual stresses present in shell and coil and maximum equivalent stress occurs at shell to coil junction. This can be analyzed by 'Shear Strain Energy Theory' which adds the principle stresses acting mutually at right angles to give the equivalent stresses.

$$f_c^2 = f_{s1}^2 + f_{s2}^2 + f_{c1}^2 - (f_{s1} \times f_{s2} + f_{s1} \times f_{c1} + f_{s2} \times f_{c1}) \dots (P.13)$$

It is prudent to check that this is within the allowable value.

P-3.2.5 As a steam coil may be subjected to vacuum (if steam is turned off) there are any possible combinations of pressure to vacuum in coil with pressure to vacuum in the shell and this gives 4 distinct limiting cases in the region of the coil. There are further 2 cases in the region where there is no coil corresponding to vacuum or pressure in the vessel with atmospheric pressure outside. By analyzing the stresses in details as indicated above, it is quite advisable to analyze the six conditions to determine the worst one.

P-3.2.6 Vessel Bottoms — It is extremely difficult to cut a half coil to suit a hemispherical or conical vessel bottom. Preferable methods in this regard should be decided between the manufacturer and designer. The recommended practice is to arrange pattern of coils radially like hub spokes and rim of a wheel.

P-3.2.7 One difficulty leading to failure of the bottom end of the half coil has been traced to be 'Steam Hammer' when steam is used intermittently in a batch process. The steam entering a cold vessel causes condensate to accelerate round the coils and hammer the exit branch and coil ends. For this reason the coil length should be limited to not more than three turns in series (see Fig. P.9). The exit nozzle should be swept rather than right angular and care taken in design of steam trap manifold.

P-3.3 Dimple Jackets

P-3.3.1 Dimple Plate Thickness

P 3.3.1.1 Case 1 (see Fig. P. 8) — Considering fixed plate with uniformly distributed load over the entire surface:

$$\text{Maximum stress at edge, } f_{e1} = \frac{3 \times W}{4 \pi \times t_{r1}^2} \quad \dots\dots (P.14)$$

Maximum deflection at the centre

$$y = (-) \frac{3 \times W (m^2 - 1) \times C^2}{16 \times \pi \times E \times m^2 \times t_{r1}^3} \quad \dots\dots (P.15)$$

P-3.3.1.2 Case 2 — Dimples can be assumed to act as 'stays' in flat heads and plate thickness is checked as below, both for:

- uniform spacing of staying, and
- non-uniform spacing of the staying

P-3.3.1.3 Case 2 (a) — Uniform spacing of stays (see Fig. P.7)

$$t_{r1 \text{ min}} = C \times \sqrt{\frac{p (a^2 + b^2)}{100 \times f}} \quad \dots\dots (P.16)$$

P-3.3.1.4 Case 2 (b) — Non-uniform spacing of stays (see Fig. P.7)

$$t_{r1 \text{ min}} = C \times \frac{e_1 + e_2}{2} \sqrt{\frac{p}{100 f}} \quad \dots\dots (P.17)$$

where $C = 0.40$ for welded stays.

The adopted thickness (t_1) shall include corrosion and manufacturing allowances, etc.

P-3.3.2 Dimple Connecting Welds — The weld attachment is made by fillet welds around holes, or if the thickness of the plate having hole is 5 mm or less and hole is 25 mm or less in diameter, the holes may be completely filled with weld metal. The allowable load on weld shall equal the product of the thickness of plate having the hole, the circumference or perimeter of hole, the allowable stress value in tension of the material being welded and joint efficiency of 55%. The connecting welds shall be checked for shear and tear failures.

P-4. CONSIDERATIONS OF STIFFENING EFFECT OF JACKET ELEMENTS (see Fig. P.2 to P.6)

P-4.1 Calculations for Elastic Buckling

Maximum working pressure, $P =$

$$\frac{E}{3} \left[\frac{200 \frac{l}{n^2} \times \frac{t_{rs}}{D_o}}{\left[1 + \left(\frac{2 \pi L}{\pi D_o} \right)^2 \right]^2} + \frac{200 n^2}{3} \left[1 + \left(\frac{\pi D_o}{2 n L} \right)^2 \right]^2 \right] \times \frac{m^2}{m^2 - 1} \times \frac{t_{rs}^3}{D_o^3} \times \left[\frac{1}{1 + \frac{1}{2} \left(\frac{\pi D_o}{2 n L} \right)^2} \right] \quad \dots\dots (P.18)$$

and 'n' should be chosen as a whole number $\left[n \geq 2 \geq \frac{\pi D_0}{2L} \right]$ in such a way that p is at its lowest value. 'n' is then the number of dents which may appear on the surface in the event of failure.

P-4.2 Calculation for Plastic Deformation

P-4.2.1 Case 1 — When $\frac{D_0}{L} \leq 5$, the maximum permissible working pressure is given by:

$$P = 200 \times f \times \frac{t_{rs}}{D_0} \times \frac{1}{1 + \frac{1.5U \left(1 - \frac{0.2 D_0}{L} \right)}{100 t_{rs}/D_0}} \dots (P.19)$$

where U = out of roundness factor.

$$a) \text{ For ovality } U = 2 \times \frac{D_{1 \max} - D_{1 \min}}{D_{1 \max} + D_{1 \min}} \times 100$$

$$b) \text{ For flatness } U = \frac{4}{D_0} \times q \times 100 \text{ (see Fig. P.4).}$$

P-4.2.2 Case 2 — When $\frac{D_0}{L} > 5$

$$P = \text{Greater of } \frac{200 \times f \times t_{rs}}{D_0} \text{ or } \frac{300 f \times t_{rs}^{2*}}{L^2} \dots (P.20)$$

P-4.2.3 In vacuum vessels the stiffening effect of the jacket elements (half pipe coils, heater channels, etc) may be taken into consideration in calculating the whole shell. For elastic buckling analysis the permissible pressure will then be greater, in proportion to the moment of inertia (taken about the axis through the appropriate centre of gravity X-X, or X'-X') and for plastic deformation analysis the permissible pressure will be greater in proportion to the cross-sectional areas with and without the jacketing elements.

P-5. FABRICATION

P-5.1 Fabrication of vessels shall be in accordance with applicable parts of the code.

P-5.2 This part covers fabrication of jacketed vessels by welding. Other methods of fabrication are permitted provided the requirements of applicable parts of this section are met.

P-5.3 Where only the inner vessel is subject to lethal service the requirements regarding radiography, post weld heat treatment shall apply only to the welds in the inner vessel and those welds attaching the jacket to the inner vessel need not be radiographed and may be fillet-welded. Post weld heat treatment shall be as required in 6.12 of the code.

P-5.4 Proper testing for pinholes in the welds is mandatory.

P-5.5 Any radial welds in closure members shall be butt-welded joints penetrating through the full thickness of the member and shall be ground flush where attachment welds are to be made.

P-6. APPLICATIONS

P-6.1 Conventional jackets are used where the internal pressure of the vessel is more than twice the jacket pressure, the jacket pressure being limited to 7 kgf/cm².

P-6.2 Dimple jackets and half-pipe coil jackets are used for vessels beyond 2m³ capacity and where the jacket pressure is the controlling factor (that is, the vessel internal pressure is less than 1.67 times the jacket pressure) in determining the vessel wall thickness. These are used for high pressure (that is, dimple jackets up to 21 kgf/cm² and half pipe coil jackets up to 52.5 kgf/cm²) and high temperature applications.

NOTE — For high temperature applications thermal expansion differentials should be considered between the metals used in the vessel and the jacket, and difference in thicknesses between the vessel and the jacket walls, and for temperatures beyond 300°C the jacket be fabricated from a metal having the same coefficient of expansion as that used for the inner vessel.

*This formula is particularly applicable when distance between stiffeners is small.